



**JOMO KENYATTA UNIVERSITY
OF
AGRICULTURE & TECHNOLOGY**

SCHOOL OF OPEN, DISTANCE AND eLEARNING

P.O. Box 62000, 00200

Nairobi, Kenya

E-mail: elearning@jkuat.ac.ke

BIT 2223 Mobile Computing

LAST REVISION ON October 29, 2013





This presentation is intended to be covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green **Exercise** tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

Errors and omissions in these notes are entirely the responsibility of the author who should only be contacted through the **Department of Curricula & Delivery (SODeL)** and suggested corrections may be e-mailed to elarning@jkuat.ac.ke.



LESSON 10

Wireless Sensor Networks

Objectives

By the end of this topic you should be able to;

- *Understand the concept behind wireless sensor networks*
- *Understand the characteristics of wireless sensor networks*
- *Explain the software and hardware needs of wireless sensor networks*
- *Explain the challenges and design principles of Wireless sensor networks*



10.1. WIRELESS SENSOR NETWORK

A Wireless Sensor Network (WSN) is a collection of sensing devices that can communicate wirelessly where each device can sense, process, and talk to its peers. It is typically, centralized in a collection point (sink or base station). It consists of spatially distributed autonomous devices using sensors to cooperatively monitor physical or environmental conditions, such as temperature, sound, vibration, pressure, motion or pollutants, at different locations.

Networked micro-sensors technology is a key technology for the future. Cheap, smart devices with multiple onboard sensors, networked through wireless links and the Internet and deployed in large numbers, provide unprecedented opportunities for instrumenting and controlling homes, cities, and the environment.



BIT 2223 Mobile Computing

In addition, networked micro-sensors provide the technology for a broad spectrum of systems in the defense arena, generating new capabilities for reconnaissance and surveillance as well as other tactical applications.

Smart disposable micro-sensors can be deployed on the ground, in the air, under water, on bodies, in vehicles, and inside buildings. A system of networked sensors can detect and track threats (e.g., winged and wheeled vehicles, personnel, chemical and biological agents) and be used for weapon targeting and area denial. Each sensor node will have embedded processing capability, and will potentially have multiple onboard sensors, operating in the acoustic, seismic, infrared (IR), and magnetic modes, as well as imagers and micro-radars. Also on-board will be storage, wireless links to neighboring nodes, and location and positioning



BIT 2223 Mobile Computing

knowledge through the global positioning system (GPS) or local positioning algorithms.

Current and potential applications of sensor networks include: military sensing, physical security, air traffic control, traffic surveillance, video surveillance, industrial and manufacturing automation, distributed robotics, environment monitoring, and building and structures monitoring. The sensors in these applications may be small or large, and the networks may be wired or wireless. However, ubiquitous wireless networks of micro-sensors probably offer the most potential in changing the world of sensing.

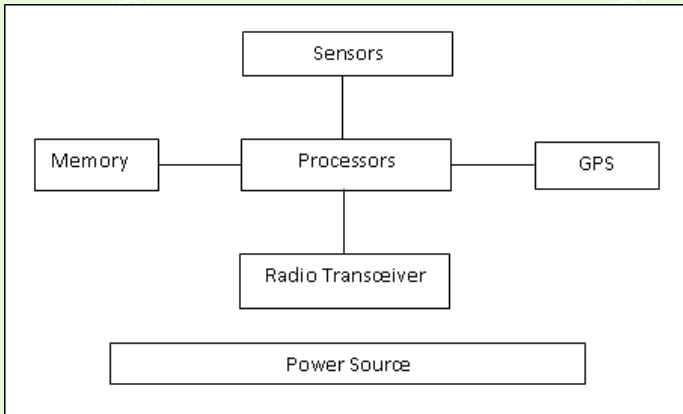


10.1.1. Wireless Sensor Network Nodes

Sensor nodes can be imagined as small computers, extremely basic in terms of their interfaces and their components. They usually consist of a processing unit with limited computational power and limited memory, sensors (including specific conditioning circuitry), a communication device (usually radio transceivers or alternatively optical), and a power source usually in the form of a battery. Other possible inclusions are energy harvesting modules, secondary ASICs, and possibly secondary communication devices (e.g. RS-232 or USB).



- Basic components of a WSN Node



10.1.2. Characteristics of a Wireless Sensor Network

- Have limited power that they can harvest or store
- Ability to withstand harsh environmental conditions



BIT 2223 Mobile Computing

- Ability to cope with node failures
- They provide a bridge between the real physical and virtual worlds
- Have ability to observe the previously unobservable at a fine resolution over large spatio-temporal scales
- They have a wide range of potential applications to industry, science, transportation, civil infrastructure, and security.
- Have a dynamic network topology
- Communication failures
- Large scale of deployment
- Unattended operation
- Energy-efficient, to maximize lifetime



BIT 2223 Mobile Computing

- Small, for ease of deployment
- Inexpensive, so that many nodes can be deployed
- Reliable, to minimize maintenance

- **Hardware**

The main challenge is to produce low cost and tiny sensor nodes. With respect to these objectives, current sensor nodes are mainly prototypes. Miniaturization and low cost are understood to follow from recent and future progress in the fields of MEMS and NEMS. Some of the nodes are still in research stage.

- **Software**

Energy is the scarcest resource of WSN nodes, and it determines the lifetime of WSNs. WSNs are meant to be deployed in large



BIT 2223 Mobile Computing

numbers in various environments, including remote and hostile regions, with ad-hoc communications as key. For this reason, algorithms and protocols need to address the following issues:

- Lifetime maximization
- Robustness and fault tolerance
- Self-configuration

10.1.3. Challenges and design principles

Military applications such as target detection, battlefield surveillance, and counter terrorism originally motivated sensor network applications. However, their advantages over traditional networks resulted in many other potential applications, ranging from infrastructure security to industrial sensing—for example, environment and habitat monitoring, healthcare applications,



home automation, and traffic control.

The design and development of a successful middleware layer must address many challenges dictated by WSN characteristics on one hand and the applications on the other.

- Distributed processing: Need algorithms that are not centralized, i.e, do not require all of the data
- Low bandwidth communication: Efficiently moves large amount of sensor data for processing
- Large scale coordination: Many independent sensor nodes need to act in concert with one another
- Real-time computational: New data is always coming, so it must be processed faster than it is generated
- Scalability



- Heterogeneity
- Systematic Design
- Privacy and Security

10.1.4. Managing limited power and resources

The advent in microelectronics technology made it possible to design miniaturized devices on the order of one cubic centimeter. Limited in energy and individual resources (such as CPU and memory), these tiny devices could be deployed in hundreds or even thousands in harsh and hostile environments. In cases where physical contact for replacement or maintenance is impossible, wireless media is the only way for remote accessibility. Hence, middleware should provide mechanisms for efficient processor and memory use while enabling lower-power commu-



BIT 2223 Mobile Computing

nication. A sensor node should accomplish its three basic operations—sensing, data processing, and communication—without exhausting resources. In energy-aware middleware, for example, most of the device's components (including the radio) are likely turned off most of the time depending on the application.

10.1.5. Scalability, mobility, and dynamic network topology

Scalability is defined as follows: if an application grows, the network should be flexible enough to allow this growth anywhere and anytime without affecting network performance. Efficient middleware services must be capable of maintaining acceptable performance levels as the network grows. Network topology is subject to frequent changes owing to factors such as malfunc-



tioning, device failure, moving obstacles, mobility, and interference. Middleware should support sensor networks' robust operation despite these dynamics by adapting to the changing network environment. Middleware also should support mechanisms for fault tolerance and sensor node self-configuration and self-maintenance.

- **Heterogeneity**

Middleware should provide low-level programming models to meet the major challenge of bridging the gap between hardware technology's raw potential and the necessary broad activities such as reconfiguration, execution, and communication. It should establish system mechanisms that interface to the various types of hardware and networks, supported only by distributed,



primitive operating-system abstractions.

- **Dynamic network organization**

Unlike traditional networks, sensor networks must deal with resources that are dynamic, such as energy, bandwidth, and processing power. Sensor networks also must support long-running applications, so routing protocols must be efficiently designed to enable the network to run as long as possible. Because knowledge of the network is essential for it to operate properly, the middleware should provide ad hoc network resource discovery. A sensor node needs to know its location in the network and in the whole network topology. In some cases, self-location by GPS is impossible, unfeasible, or expensive. Important system parameter issues, such as network size and density per square



mile, affect the trade-offs among latency, reliability, and energy.

- **Real-world integration**

Most sensor network applications are real-time phenomena, where time and space are extremely important. Hence, middleware should provide real-time services to adapt to the changes and provide consistent data.

- **Application knowledge**

Application knowledge's design principles dictate another important and unique property of WSN middleware. Middleware must include mechanisms for injecting application knowledge of WSN's infrastructure. This lets developers map application communication requirements to network parameters, which enable them to fine-tune network monitoring. Much existing mid-



Middleware is coupled to specific applications. However, middleware is intended to support a wide range of applications. So, developers must explore the trade-offs between the degree of application specificity and middleware generality.

- **Data aggregation**

Most sensor network applications involve nodes that contain redundant data and are located in a specific local region. These traits open the possibility for in-network aggregation of data from different sources, eliminating redundancy and minimizing the number of transmissions to the sink. This aggregation saves considerable energy and resources, given that communications cost is much higher than computation costs. This paradigm shifts the focus from the traditional address-centric approaches



for networking to a more data-centric approach.

- **Quality of service**

Quality of service is an overused term with multiple meanings and perspectives from different research and technical communities. In WSNs, we can view QoS from two perspectives: application-specific and network. The former refers to QoS parameters specific to the application, such as sensor node measurement, deployment, and coverage and number of active sensor nodes. The latter refers to how the supporting communication network can meet application needs while efficiently using network resources such as bandwidth and power consumption.

Traditional QoS mechanisms used in wired networks aren't adequate for WSNs because of constraints such as resource lim-



itations and dynamic topology. Therefore, middleware should provide new mechanisms to maintain QoS over an extended period and even adjust itself when the required QoS and the state of the application changes. Middleware should be designed based on trade-offs among performance metrics such as network capacity or throughput, data delivery delay, and energy consumption.

- **Security**

WSNs are being widely deployed in domains that involve sensitive information—for example, healthcare and rescue. The untethered and large deployment of WSNs in harsh environments increases their exposure to malicious intrusions and attacks such as denial of service. In addition, the wireless medium facili-



BIT 2223 Mobile Computing

tates eavesdropping and adversarial packet injection to compromise the network's functioning. All these factors make security extremely important. Furthermore, sensor nodes have limited power and processing resources, so standard security mechanisms, which are heavy in weight and resource consumption, are unsuitable. These challenges increase the need to develop comprehensive and secure solutions that achieve wider protection, while maintaining desirable network performance. Middleware efforts should concentrate on developing and integrating security in the initial phases of software design, hence achieving different security requirements such as confidentiality, authentication, integrity, freshness, and availability.



10.1.6. Fields of application of wireless sensor networks

There are numerous different fields of application of sensor networks. For example, forest fires can be detected by sensor networks so that they can be fought at an early stage. Sensor networks can be used to monitor the structural integrity of civil structures by localizing damage for example in bridges. Further, they are used in the health care sector to monitor human physiological data. The following sections outline selected applications of wireless sensor networks.

- Entertainment
- Transportation and logistics
- Smart grids and energy control systems
- Habitat and Ecosystem Monitoring



BIT 2223 Mobile Computing

- Seismic Monitoring
- Civil Structural Health Monitoring
- Monitoring Groundwater Contamination
- Rapid Emergency Response
- Industrial Process Monitoring
- Perimeter Security and Surveillance
- Automated Building Climate Control



Revision Questions

EXERCISE 1.  Explain the software and hardware needs of wireless sensor networks

Example  . Expound on Perimeter Security and Surveillance in regard to sensor networking

Solution: Revision ☐

EXERCISE 2.  Explain the challenges and design principles of Wireless sensor networks



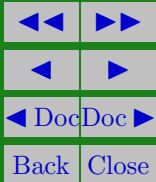
JKUAT SODeL
©2013

BIT 2223 Mobile Computing

Solutions to Exercises

Exercise 1. Revision

Exercise 1





JKUAT SODeL
©2013

